

10th grade – Term 3 Practice Activities: Pedagogical Summary (C0–C7)

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1 G10_T3_C0_practice_activity.tex

This file develops foundational understanding of continuous probability distributions and the transition from discrete-style thinking to density-based reasoning by first reinforcing PMF structure, then pushing students to organize event probabilities systematically from tabular models. The criterion focus is on validating a probability model (normalization and nonnegativity), computing event probabilities, and interpreting results in context. Students are expected to develop careful additive reasoning over supports, gradually moving from direct lookup to multi-parameter and constrained PMF reasoning.

1. **App Notifications per Hour:** Introductory PMF validation with one unknown parameter, followed by exact, cumulative, and interval-event probabilities.
2. **Defective Items in a Batch:** Reinforces the same structure in a quality-control context, emphasizing translation from table to event sums.
3. **Lost Pens in a Small Box:** Early comparative practice on complementary event forms (e.g., at least vs at most) with simple support values.
4. **Late Deliveries in One Day:** Contextual counting model that strengthens event selection for threshold-based probabilities.
5. **Customers in a 10-Minute Window:** Introduces symbolic parameter notation inside the PMF and consolidates normalization-to-parameter solving.

6. **Customers in a Short Checkout Window:** Similar structure with altered support and values, promoting robustness rather than pattern memorization.
7. **Correct Answers on a Short Quiz:** Moves to piecewise PMF definitions (in terms of a scalar parameter), encouraging algebraic organization before computation.
8. **Messages Received in an Hour:** Coefficient-pattern PMF (e.g., multiples of a base parameter) that trains proportional reasoning in discrete distributions.
9. **Calls Received in a Busy Hour:** Mixed fixed-and-parameter entries requiring students to coordinate substitution and aggregation accurately.
10. **Goals in a Match Summary:** Symbolic PMF with fractional expressions, increasing algebraic sophistication while preserving event-sum interpretation.
11. **Late Buses in a Morning:** Two-parameter PMF setup that builds constraint management from normalization plus additional relationships.
12. **Deliveries Delayed in a Morning:** Similar dual-parameter reasoning in another context to reinforce solving coupled linear conditions.
13. **Customers in a Short Window:** Five-value support with parameterized pattern, supporting more complex cumulative-event calculations.
14. **Correct Answers on a 3-Question Check:** Compact support with symbolic entries to sharpen exact-event selection and bounded-interval sums.
15. **Scores in a Quick Check:** Comparative symbolic setup that emphasizes model validity checks before probability queries.
16. **Yellow Cards in a Match:** Broader support and context interpretation, reinforcing probability statements as percentage outcomes.
17. **Defects Found in Final Check:** Piecewise symbolic PMF with embedded constraint terms, requiring more advanced consistency checking.
18. **Phone Calls to the Office:** Parameterized frequency model that integrates table reading with event-logic precision.
19. **Homework Submissions Missing:** School-based context for multi-value event unions and complement-based shortcuts.
20. **Returns Processed at a Counter:** Capstone-style PMF exercise combining parameter solving, event construction, and contextual interpretation.

2 G10_T3_C1_practice_activity.tex

This file focuses on identifying and validating continuous PDFs, especially uniform and linear models, using support, nonnegativity, and total area. Its learning purpose is to establish validity criteria geometrically and algebraically before any advanced integration techniques. Students are expected to develop geometric reasoning about area, model-checking discipline, and interpretation of how density shape redistributes likelihood across the support.

1. **Checking Two Candidate Density Functions I:** Introductory comparison of two constant models, deciding validity by rectangle area and then computing a basic interval probability.
2. **Checking Two Candidate Density Functions II:** Comparative linear-increasing candidates, using triangle area to validate one model and reject another.
3. **Checking Two Candidate Density Functions III:** Decreasing linear candidates on shifted support, empha-

sizing trapezium-area reasoning.

4. **Water Storage Level (Uniform):** Contextual uniform PDF with unknown height, solving for the constant from total area.
5. **Startup Delay (Linear, Endpoint at 0):** First pure endpoint-zero linear model with scale parameter determination.
6. **Packing Score (Linear, Both Endpoints Nonzero):** Shifted linear form on nonzero interval, strengthening affine-density normalization.
7. **Classroom Temperature Band (Uniform):** Uniform model on an interval crossing zero to reinforce support handling.
8. **Assembly Time (Linear, Endpoint at 0):** Decreasing linear density that contrasts with earlier increasing cases.
9. **Quality Index (Linear, Both Endpoints Nonzero):** Another shifted linear model for fluent validity checks under translation.
10. **Rainfall Amount (Uniform):** Model with unknown support endpoint; students infer domain width from known density.
11. **Battery Discharge (Linear, Endpoint at 0):** Linear model with variable endpoint parameter, integrating support and slope constraints.
12. **Delivery Score (Linear, Both Endpoints Nonzero):** Mixed constant-plus-linear expression, advancing symbolic normalization skills.
13. **Clinic Waiting Time (Uniform):** Endpoint-density conditions used to recover a support limit and validate consistency.
14. **Task Completion Fraction (Linear, Endpoint at 0):** Uses boundary values to build a line model and then verify area 1.
15. **Final Performance Measure (Linear, Both Endpoints Nonzero):** Structured endpoint-value reconstruction of a linear PDF over a variable interval.
16. **Sensor Error Level (Uniform with Variable Domain):** Fully symbolic uniform case, abstracting the validity logic.
17. **Traffic Flow Index (Linear with Variable Domain):** Symbolic increasing linear PDF on $[0, b]$, emphasizing formula generalization.
18. **Machine Output Deviation (Linear with One Variable Endpoint):** Shifted-origin linear density with variable right endpoint.
19. **Inspection Time (Uniform with Two Variable Endpoints):** Two-parameter support width case, highlighting dependence of height on interval length.
20. **Load Measure (Increasing Linear with Two Variable Endpoints):** Symbolic slope-and-width interplay in a translated coordinate frame.
21. **Cooling Level (Decreasing Linear with Two Variable Endpoints):** Mirror case of increasing models to reinforce geometric duality.
22. **Delivery Deviation (Shifted Linear with Two Variable Endpoints):** Capstone symbolic linear-PDF validation with shifted variable structure.

3 G10_T3_C2_practice_activity.tex

This file focuses on interpreting probability as area under the curve in uniform and piecewise uniform distributions. The learning purpose is to consolidate area-as-probability thinking across fixed, variable, and parameterized supports, including complements and union structures. Students are expected to develop interval-partition reasoning, symbolic width-based calculation, and structured decomposition of piecewise regions.

1. **Introductory Uniform Distribution:** Foundational uniform model on a fixed interval with direct interval, tail, and complement probabilities.
2. **Introductory Uniform Distribution with Variable Endpoint:** Repeats core tasks symbolically when the upper support boundary is unknown.
3. **Introductory Uniform Distribution with Variable Startpoint:** Parallel symbolic case where the lower boundary varies.
4. **Introductory Uniform Distribution with Variable Endpoints:** Fully general $\text{Uniform}(a, b)$ reasoning with multiple query types.
5. **Two-Interval Piecewise Uniform (Basic) [0 to 10]:** First piecewise constant density; students add areas across intervals of different heights.
6. **Two-Interval Piecewise Uniform (Basic) [-2 to 6]:** Same structure on shifted support to reinforce coordinate flexibility.
7. **Two-Interval Piecewise Uniform with Unique Variable Endpoint:** One branch has fixed height and one fixed level with unknown end, requiring support recovery.
8. **Two-Interval Piecewise Uniform with Unique Variable Startpoint:** Mirror setup with unknown startpoint inferred from validity.
9. **Two-Interval Piecewise Uniform with Variable Endpoint:** Height depends on endpoint parameter, linking normalization with expression manipulation.
10. **Two-Interval Piecewise Uniform with Variable Start:** Analogous start-parameter model for generalized symbolic probability formulas.
11. **Three-Interval Piecewise Uniform (Combining Regions):** Introduces multi-block area composition for nontrivial interval events.
12. **Piecewise Uniform with Complement Reasoning:** Emphasizes efficient use of complement events over direct multi-piece addition.
13. **Piecewise Uniform with Unknown k :** Determines common scale factor first, then evaluates interval probabilities.
14. **Piecewise Uniform with Unknown k and Validity Check:** Adds explicit model-validity verification before event calculations.
15. **Four-Interval Piecewise Uniform (Organizing Across Many Intervals):** Trains careful bookkeeping when an event spans many segments.
16. **Structured Piecewise Uniform with Parameter k :** Hybrid fixed-and-parameter heights supporting staged solving.
17. **Symbolic Endpoint (Conceptual Effect of Width):** Focuses on how scaling interval width changes density height and probabilities.
18. **Infer Missing Endpoint from a Probability Condition:** Inverse-style task: recover model parameters from a given event probability.
19. **Rich Piecewise Uniform Design Problem:** Capstone modeling exercise combining unknown heights, multiple

intervals, and consistency constraints.

4 G10_T3_C3_practice_activity.tex

This file focuses on piecewise linear and triangular-type density graphs, emphasizing a central most probable region and geometric probability reasoning. The learning purpose is to move from simple linear PDFs to richer peaked and asymmetric constructions while preserving area-based validity and event calculation methods. Students are expected to develop geometric decomposition skills, central-region interpretation, parameter solving, and comparative reasoning across families of linear densities.

1. **Problem 1 (Introductory Linear Density):** Single rising line on fixed support; validates PDF by triangle area and computes a starter subinterval probability.
2. **Problem 2 (Basic Symmetric Triangular Density):** Symmetric triangle model with complement-based central-band probability.
3. **Problem 3 (Basic Asymmetric Triangular Density):** Asymmetric peak case showing how unequal side widths affect tail areas.
4. **Problem 4 (Triangular Density with Split-Interval Areas):** Practices splitting a target event across both sides of the peak.
5. **Problem 5 (Richer Peaked Piecewise Linear Shape):** Introduces non-symmetric piecewise slopes and multi-region area composition.
6. **Problem 6 (Triangular PDF with Unknown Scale k):** Parameter determination through total area followed by event computation.
7. **Problem 7 (Asymmetric Triangular PDF with Unknown k):** Extends parameter solving to unequal left/right linear rules.
8. **Problem 8 (Three-Segment Peaked Density with Parameter):** More advanced piecewise linear model with three segments and symbolic normalization.
9. **Problem 9 (Comparing Two Triangular-Type Densities):** Comparative reasoning task on shape effects for central and tail probabilities.
10. **Problem 10 (Variable Peak Location p):** Investigates how moving the mode changes distribution behavior and interval probabilities.
11. **Problem 11 (Symbolic Height and Asymmetry):** Uses symbolic peak height to connect geometry and algebra in probability expressions.
12. **Problem 12 (Advanced Parameterized Triangular Model):** Fully parameterized support-and-mode framework with generalized formulas.
13. **Problem 13 (First Multi-Parameter Piecewise Linear Density):** Two-parameter family requiring structured symbolic area management.
14. **Problem 14 (Multi-Parameter Comparison Across Two Settings):** Compares parameter settings within one family to interpret shifts in concentration.
15. **Problem 15 (Capstone Multi-Parameter Piecewise Linear Model):** Capstone synthesis of validation, symbolic probability, and interpretation for a general peaked model.

5 G10_T3_C4_practice_activity.tex

This file focuses on mean, variance, and standard deviation in continuous distributions such as uniform, triangular, and normal models. Its learning purpose is to connect distribution shape to center and spread, then compare models and parameter effects using these summary measures. Students are expected to develop formula fluency, interpretive reasoning about variability, and comparative judgment across contexts.

1. **Problem 1 (Streaming Time – Uniform):** Computes and interprets $E[X]$, variance, and SD for a concrete uniform model.
2. **Problem 2 (Bus Wait Time – Triangular):** Extends moment calculations to triangular distributions and links parameters to skew.
3. **Problem 3 (Quiz Scores – Normal):** Reviews mean/variance structure of normal models and sigma-based interval interpretation.
4. **Problem 4 (Comparing Uniform Distributions):** Comparative task on center and spread across multiple uniform supports.
5. **Problem 5 (Comparing Triangular Distributions):** Contrasts triangular parameter sets to analyze mean shifts and variability changes.
6. **Problem 6 (Comparing Normal Distributions):** Distinguishes effects of changing mean versus changing standard deviation.
7. **Problem 7 (Uniform with Parameters):** Symbolic uniform model deriving moments in parameter form.
8. **Problem 8 (Triangular with Parameters):** Parameterized triangular moments and shape interpretation.
9. **Problem 9 (Normal with Parameters):** Abstract normal model emphasizing direct mapping from (μ, σ^2) notation.
10. **Problem 10 (Uniform with Parameters):** Symmetric-around-center uniform form linking radius parameter to spread.
11. **Problem 11 (Triangular with Parameters):** Shifted triangular family showing translation effects on mean.
12. **Problem 12 (Normal with Parameters):** Parameterized reaction-time model interpreting constraints on spread parameters.
13. **Problem 13 (Uniform with Parameters):** Non-symmetric symbolic interval emphasizing width-driven variance.
14. **Problem 14 (Triangular with Parameters):** Scaled triangular family connecting mode placement and dispersion.
15. **Problem 15 (Normal with Parameters):** Capstone symbolic normal model combining location and scale parameters in context.

6 G10_T3_C5_practice_activity.tex

This file focuses on using the standard normal distribution and reading normal-distribution structure in context. The learning purpose is to build table fluency for $\Phi(z)$, including one-tail, central-band, union, absolute-value, and symmetry-based probabilities. Students are expected to develop precise translation between probability language and shaded-region structure on the standard normal curve.

1. **Standard Normal Table Practice IV:** Introductory table retrieval for left-tail and right-tail probabilities.
2. **Central Band with a Negative Bound:** Computes an interval spanning negative to positive z-values using

symmetry.

3. **Two-Tail Union I:** Disjoint union of lower and upper tails from two different cutoffs.
4. **Complement with Absolute Value:** Connects $|Z| \leq a$ and $|Z| \geq a$ through complement reasoning.
5. **Interval Across Zero:** Reinforces central-interval computation when endpoints lie on opposite sides of zero.
6. **Two-Tail Union II:** Additional two-tail practice with different thresholds to strengthen setup accuracy.
7. **Upper Slice and Upper Tail:** Distinguishes bounded upper intervals from full right tails.
8. **Interval with a Small Negative Bound:** Fine-grained table lookup with close-to-zero negative endpoint.
9. **Two-Tail Union III:** Consolidates disjoint-tail probability accumulation in a third variant.
10. **Symmetry Near Zero:** Uses symmetry properties to simplify calculations for near-center regions.

7 G10_T3_C6_practice_activity.tex

This file focuses on standardization of the normal variable and interpretation of z-scores in contextual situations. The learning purpose is to make $z = (x - \mu)/\sigma$ an operational tool for converting real-world thresholds and bands into standard-normal probability statements. Students are expected to develop procedural fluency in standardization, interpretive fluency for z-scores, and flexibility with tails, bands, and unions.

1. **Exam Performance Benchmark:** Right-tail contextual probability after standardizing a score threshold.
2. **Delivery Time Reliability:** Left-tail probability with negative z-score and symmetry use.
3. **Apple Mass Acceptance Band:** Two-sided acceptance interval converted to a central z-band.
4. **Commute Duration Window:** Symmetric interval around the mean with paired z-boundaries.
5. **Battery Life Threshold:** Upper-tail event on a small-scale measurement context.
6. **Streaming Download Speed Band:** Bounded interval probability with moderate z-scores.
7. **Coffee Shop Queue Extremes:** Union of two outer regions requiring disjoint-tail addition.
8. **Daily Screen-On Time:** Everyday context for tail/interval conversion through z-scores.
9. **Daily Step Count Extremes:** Larger-unit context reinforcing standardization regardless of scale.
10. **Classroom Temperature Comfort Range:** Comfort-band probability emphasizing interpretation of central coverage.

8 G10_T3_C7_practice_activity.tex

This file focuses on inverse normal ideas and solving contextual problems where a value must be recovered from a given probability. The learning purpose is to reverse the forward-probability workflow: identify cumulative targets, obtain quantile z-values, and map back to x-scale cutoffs or intervals. Students are expected to develop quantile reasoning, tail-to-cumulative translation, and interpretation of percentile-based decisions.

1. **Top 5% Exam Scores:** Upper-tail cutoff problem translating to the 95th percentile.
2. **Bottom 10% Commute Times:** Lower-tail quantile recovery from a percentile condition.
3. **Central 95% Device Battery Life:** Two-quantile interval recovery for a middle-probability band.
4. **Bottom 20% Streaming Speeds:** Contextual lower quantile from cumulative probability 0.20.
5. **Top 2% Plant Heights:** Extreme upper-tail cutoff requiring high-percentile inverse lookup.
6. **Bottom 1% Package Delivery Times:** Extreme lower-tail quantile emphasizing rare-event thresholds.
7. **Middle 80% Fitness Run Times:** Central band from paired 10th and 90th percentiles.

8. **Top 10% Product Lifetime:** Upper-decile cutoff in reliability context.
9. **Middle 90% Manufacturing Tolerance Width:** Symmetric tolerance interval from 5th and 95th percentiles.
10. **Middle 96% Chemistry Scores:** Wider central coverage interval from matched tail allocations.